

11(a)	$eV = hf$ $f = 1.60 \times 10^{-19} \times 100\,000 / 6.63 \times 10^{-34}$	C1
	= 2.41×10^{19} Hz	A1
11(b)	(aluminium filter) absorbs (most) low energy X-rays	B1
	Any 2 from • X-ray beam contains many wavelengths • so low energy X-rays are not absorbed in the body • low energy X-rays can cause harm but do not contribute to the image	B2
11(c)(i)	$I / I_0 = e^{-\mu x}$ $e^{-0.23 \times 0.80} = 0.83$	C1
	17% absorbed	A1
11(c)(ii)	bone is seen as lighter / muscle is seen as darker	B1
	<i>either</i> bone has a higher μ value so absorbs more <i>or</i> muscle has a lower μ value so transmits more	B1